

전산유체해석실습 보고서

11월 14일

항공기계공학과

2023010605 유수연

SU2 실습 과제 NASA Case_2565

- 아래와 같이 변경한 뒤 cmd를 돌려줍니다.

```
% ----- COMPRESSIBLE FREE-STREAM DEFINITION -----%
%
% Mach number (non-dimensional, based on the free-stream values)
MACH_NUMBER= 0.8395
%
% Angle of attack (degrees, only for compressible flows)
AOA= 6.06
%
% Side-slip angle (degrees, only for compressible flows)
SIDESLIP_ANGLE= 0.0
%
% Init option to choose between Reynolds (default) or thermodynamics quantities
% for initializing the solution (REYNOLDS, TD_CONDITIONS)
INIT_OPTION= REYNOLDS
%
% Free-stream option to choose between density and temperature (default) for
% initializing the solution (TEMPERATURE_FS, DENSITY_FS)
FREESTREAM_OPTION= TEMPERATURE_FS
%
% Free-stream temperature (288.15 K by default)
FREESTREAM_TEMPERATURE= 288.15
%
% Reynolds number (non-dimensional, based on the free-stream values)
REYNOLDS_NUMBER= 11.71E6
%
% Reynolds length (1 m by default)
REYNOLDS_LENGTH= 0.64607
```



```
610| 3.3873e-01| -8.669166| -10.672112| 0.488937| 0.052449|
611| 3.3871e-01| -8.672532| -10.674535| 0.488937| 0.052449|
612| 3.3872e-01| -8.675898| -10.676952| 0.488938| 0.052449|
613| 3.3870e-01| -8.679263| -10.679364| 0.488938| 0.052449|
614| 3.3868e-01| -8.682629| -10.681771| 0.488938| 0.052449|
615| 3.3867e-01| -8.685995| -10.684172| 0.488938| 0.052449|
616| 3.3867e-01| -8.689361| -10.686567| 0.488938| 0.052449|
617| 3.3865e-01| -8.692728| -10.688956| 0.488939| 0.052449|

----- Solver Exit -----
All convergence criteria satisfied.
+-----+
| Convergence Field | Value | Criterion | Converged |
+-----+
| Cauchy[CD] | 9.95963e-07 | < 1e-06 | Yes |
+-----+

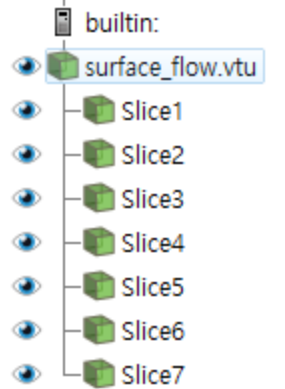
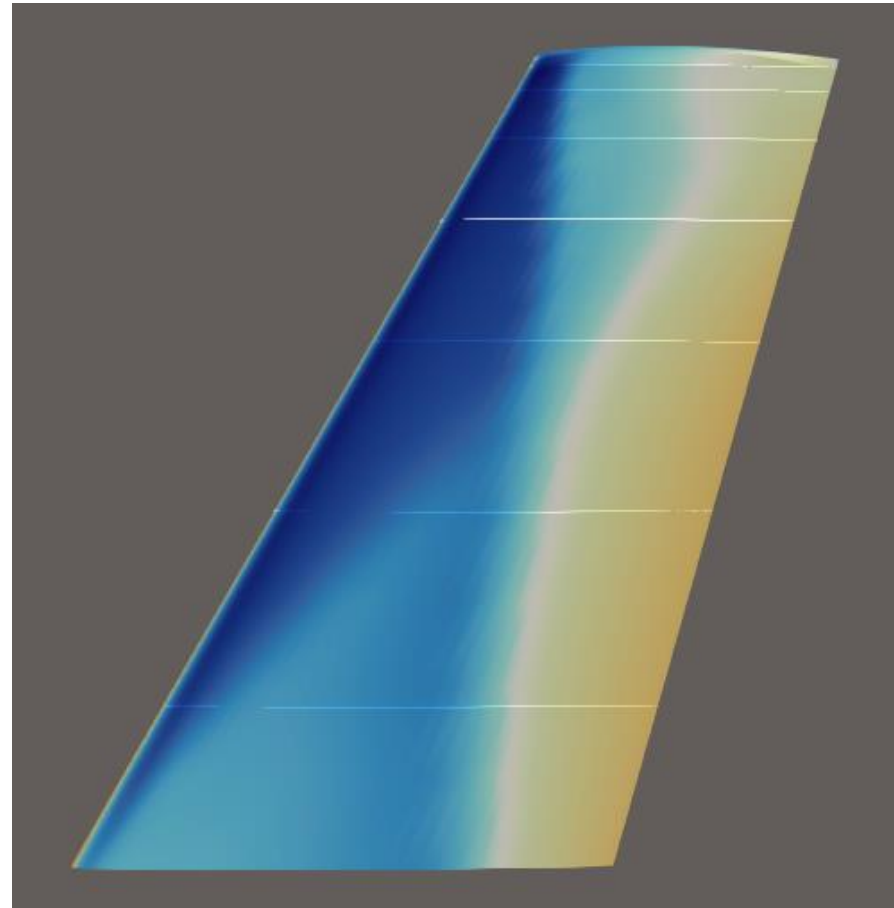
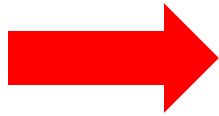
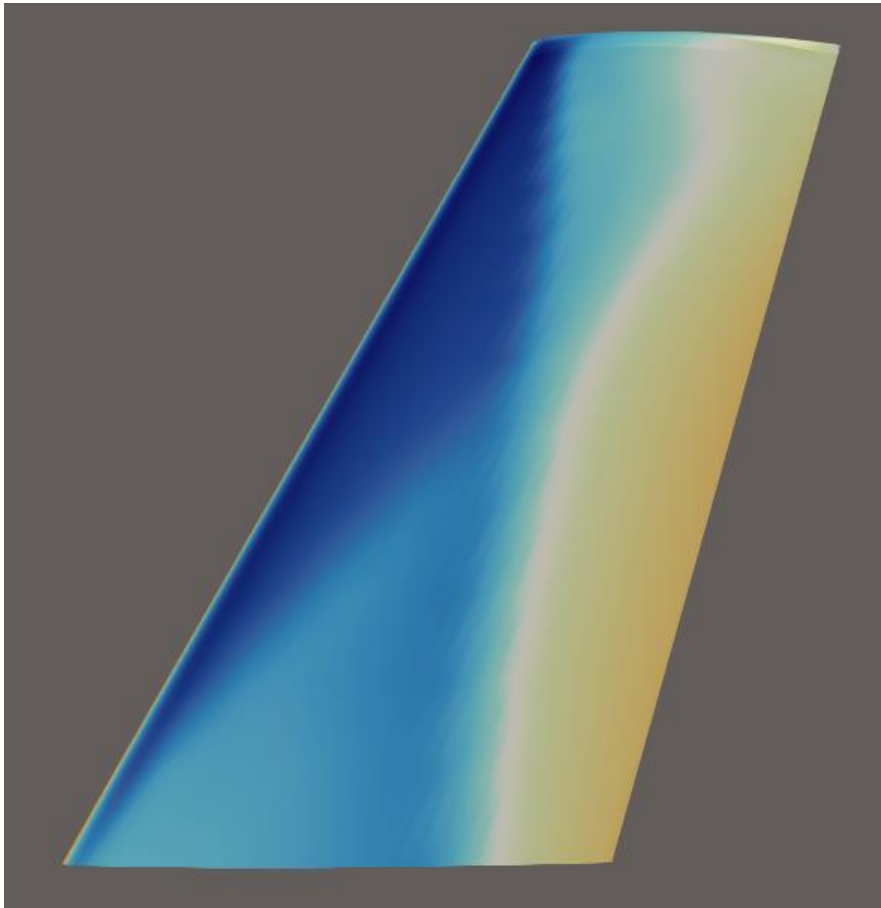
+-----+
| File Writing Summary | Filename |
+-----+
| SU2 binary restart | restart_flow.dat |
| Paraview | flow.vtu |
| Paraview surface | surface_flow.vtu |
+-----+

----- Finalizing Solver -----
Deleted CNumerics container.
Deleted CIntegration container.
Deleted CSolver container.
Deleted CIteration container.
Deleted CInterface container.
Deleted CGeometry container.
Deleted CFreeFormDefBox class.
Deleted CSurfaceMovement class.
Deleted CVolumetricMovement class.
Deleted CConfig container.
Deleted nInst container.
Deleted COutput class.

----- Exit Success (SU2_CFD) -----
```

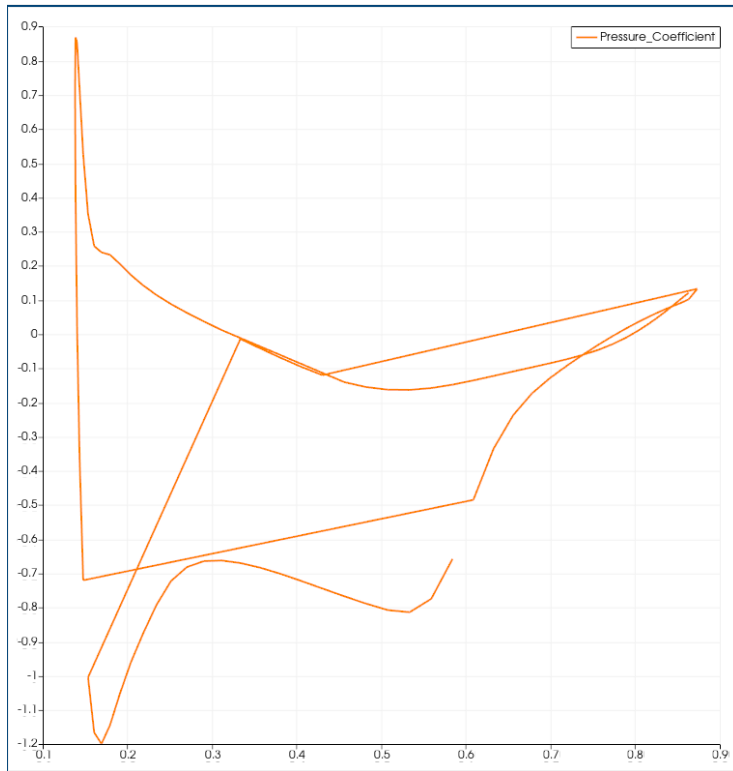
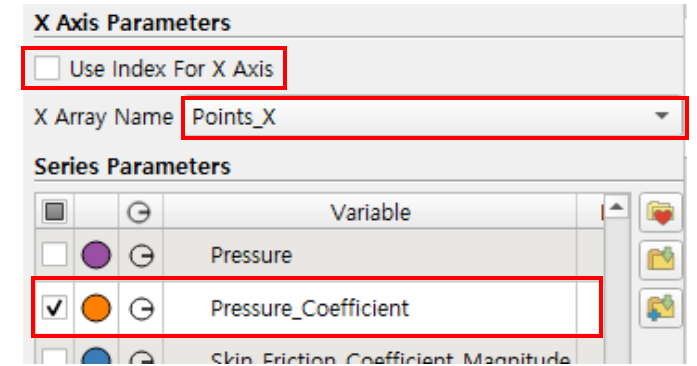
SU2 실습 과제 NASA Case_2565

- Pressure로 변경해준 뒤 Surface에서 Slice를 각각 Y-normal로
- 0.2 / 0.44 / 0.65 / 0.8 / 0.9 / 0.96 / 0.99 로 변경합니다.

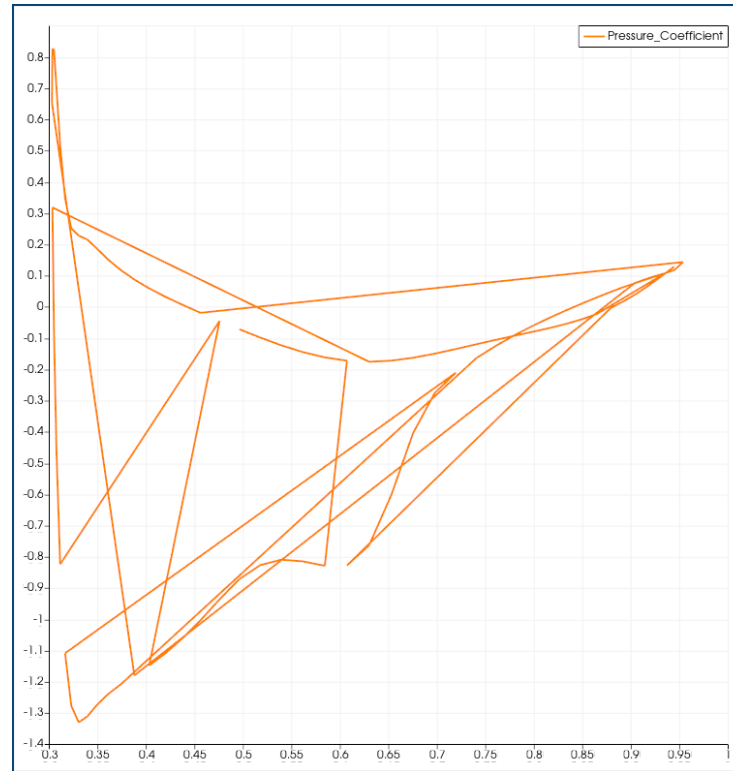


SU2 실습 과제 NASA Case_2565

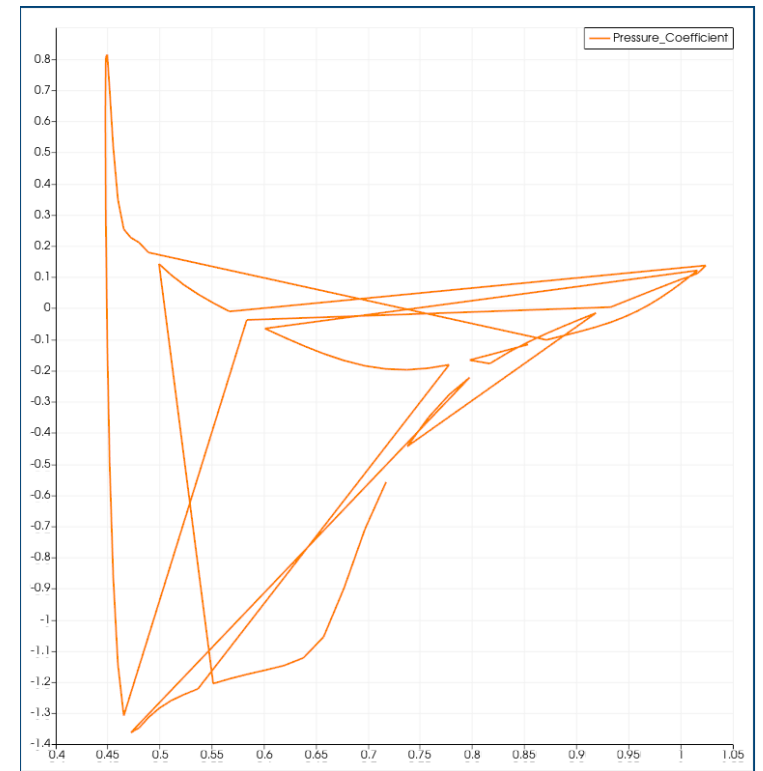
- 각 슬라이스에서 plot data를 열어 다음과 같이 설정해줍니다.



[0.2]

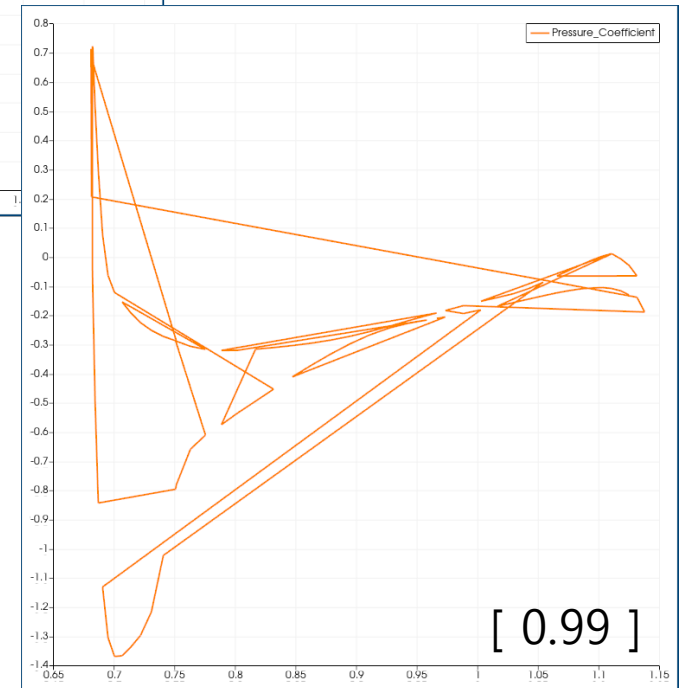
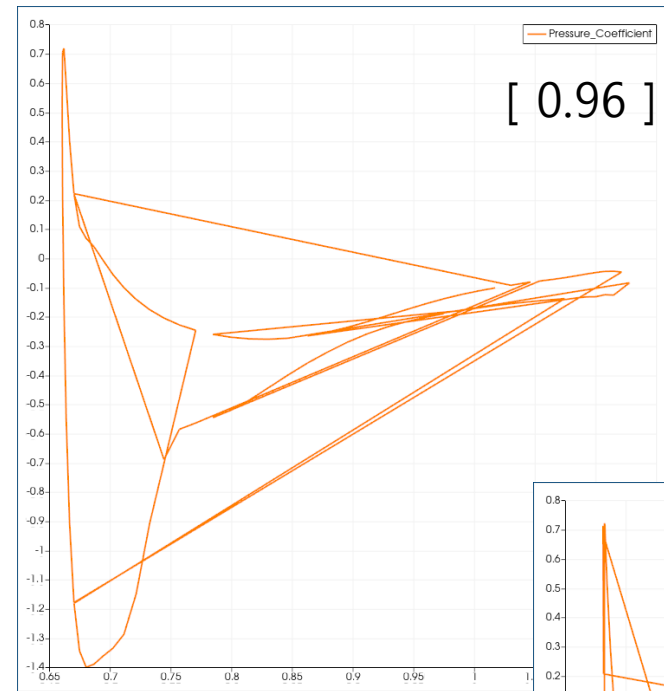
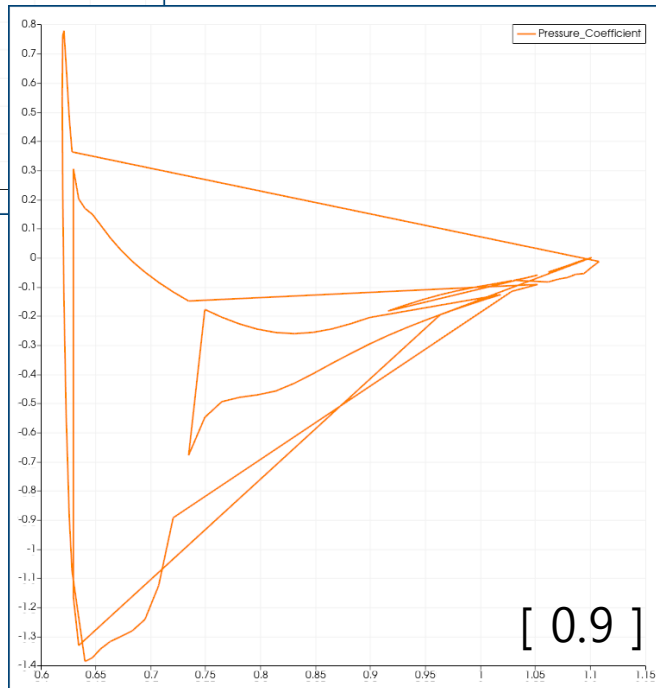
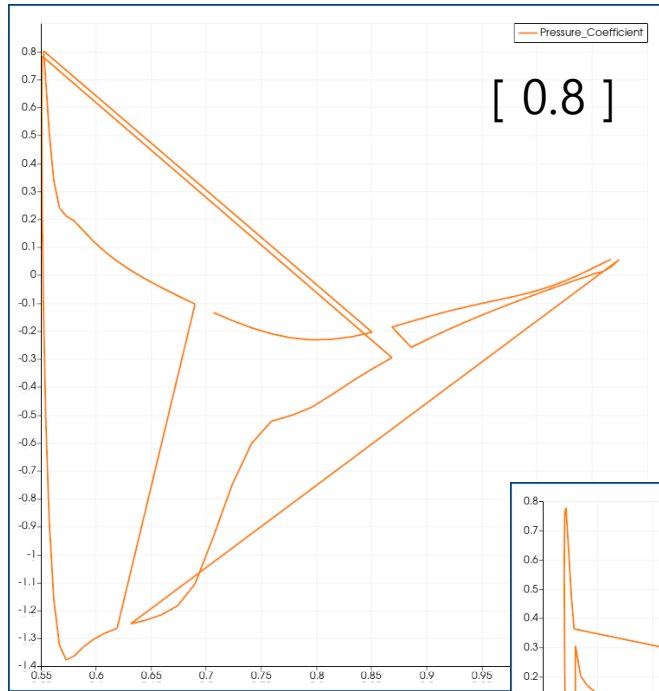


[0.44]



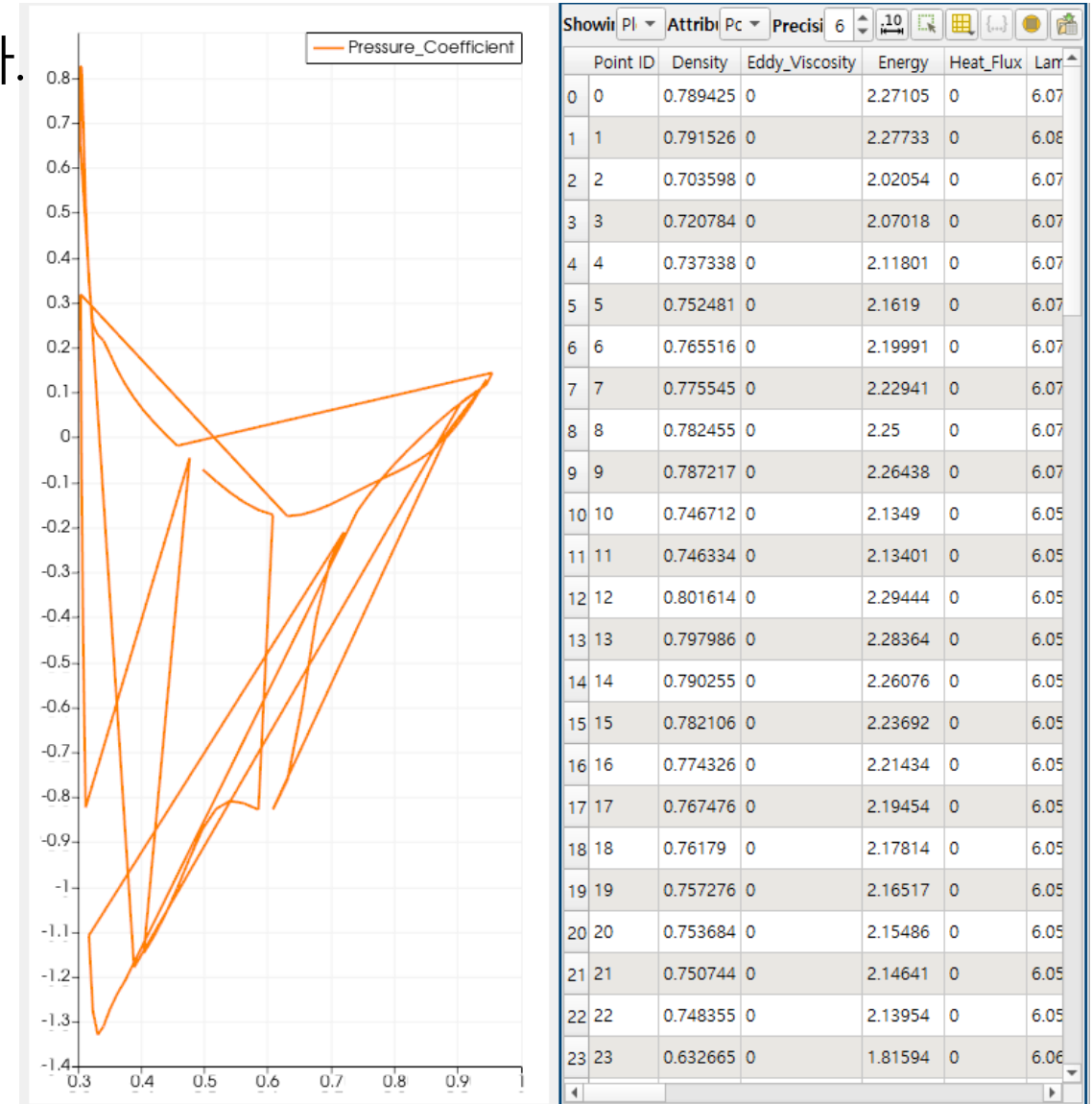
[0.65]

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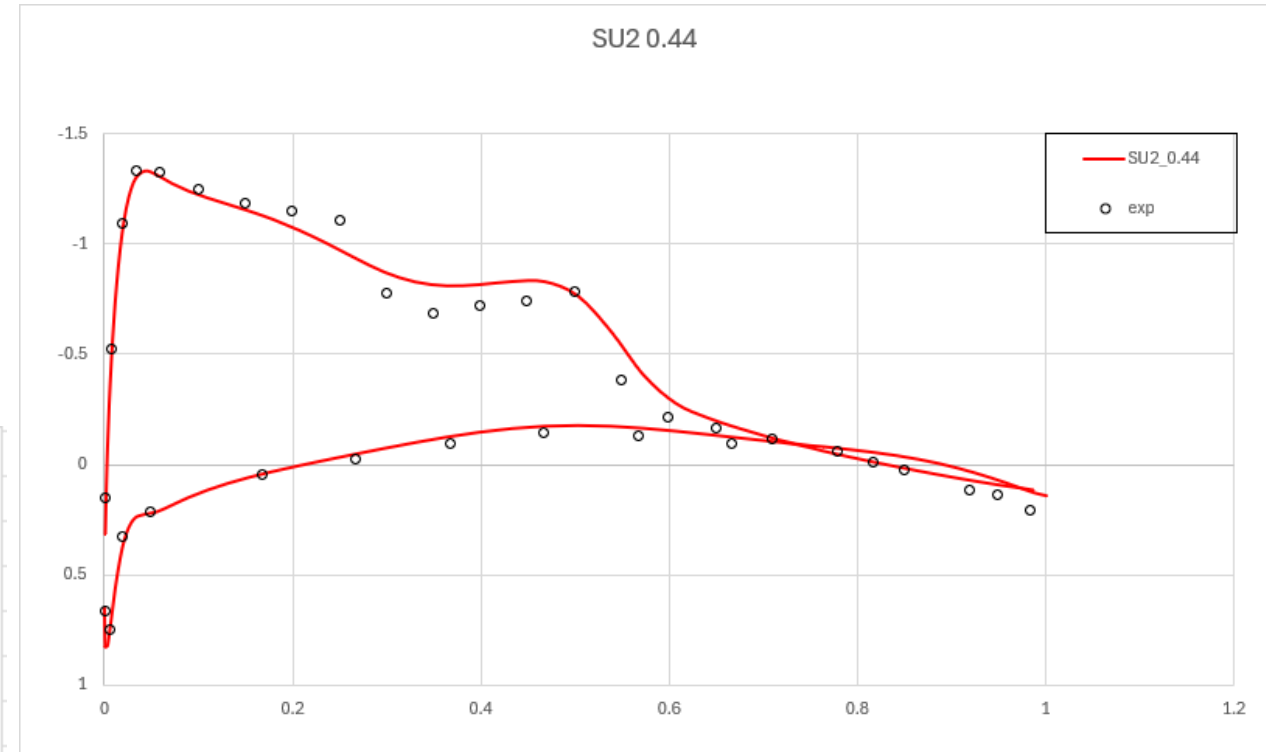
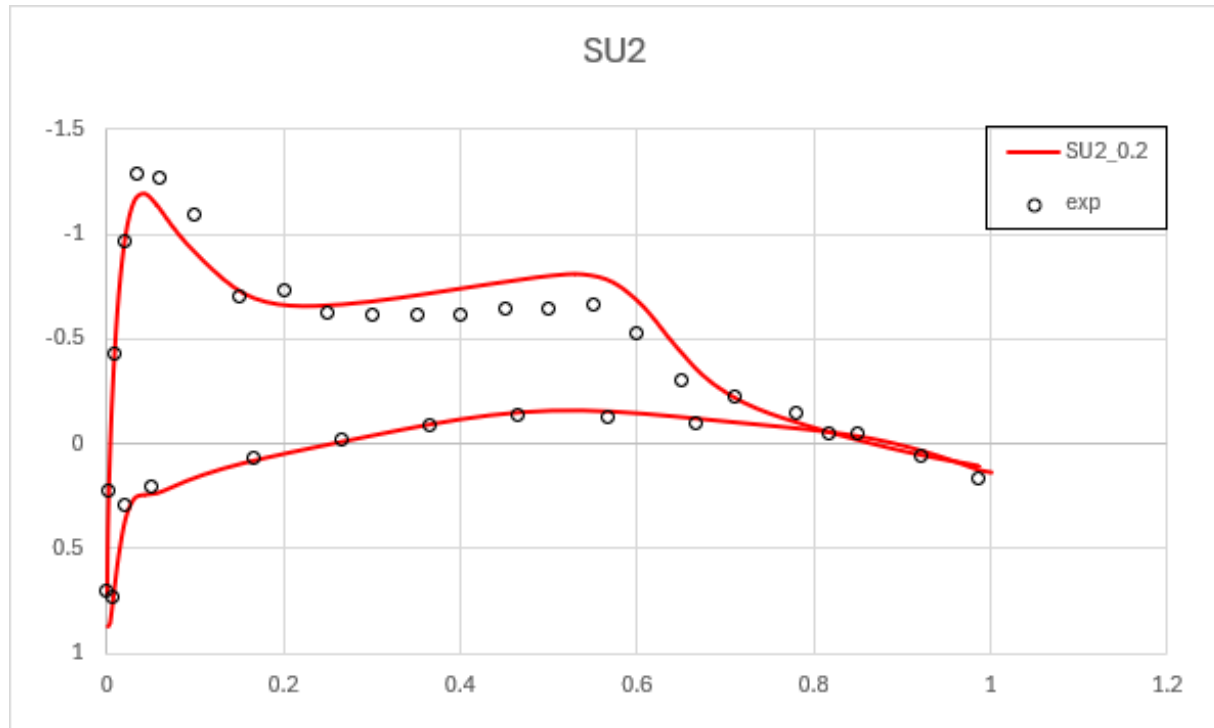


SU2 실습 과제 NASA Case_2565

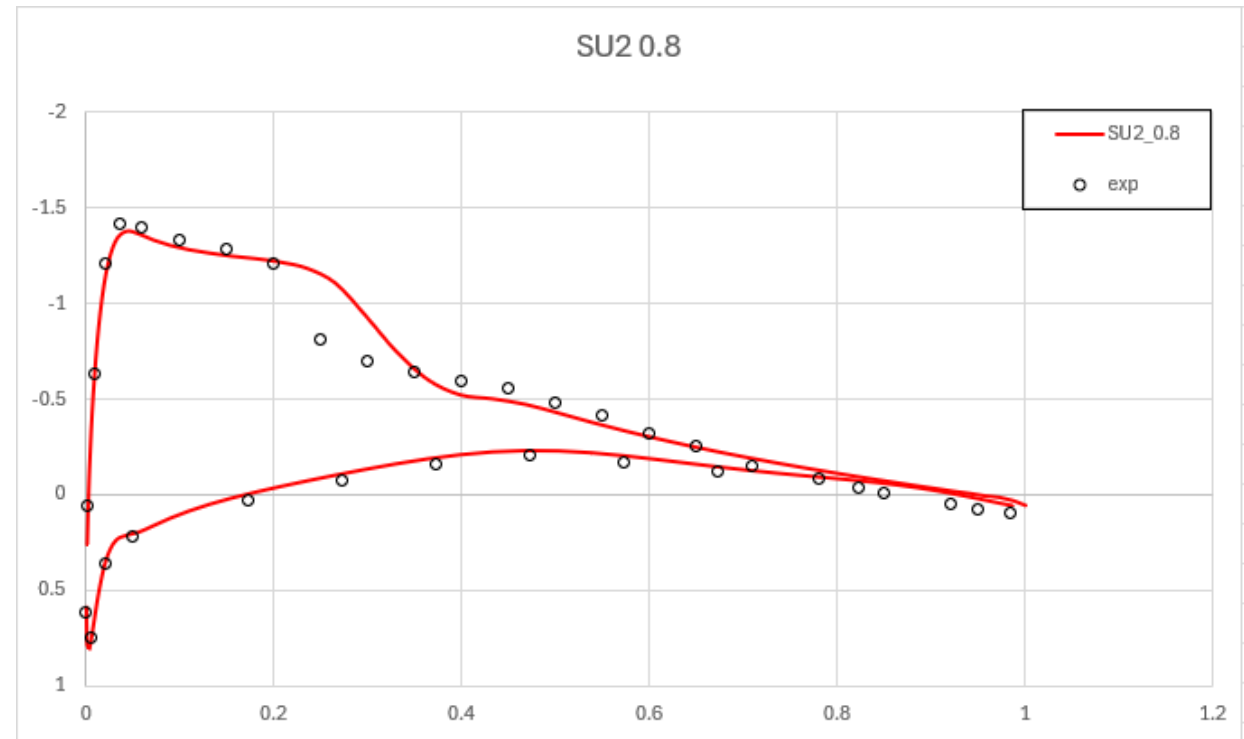
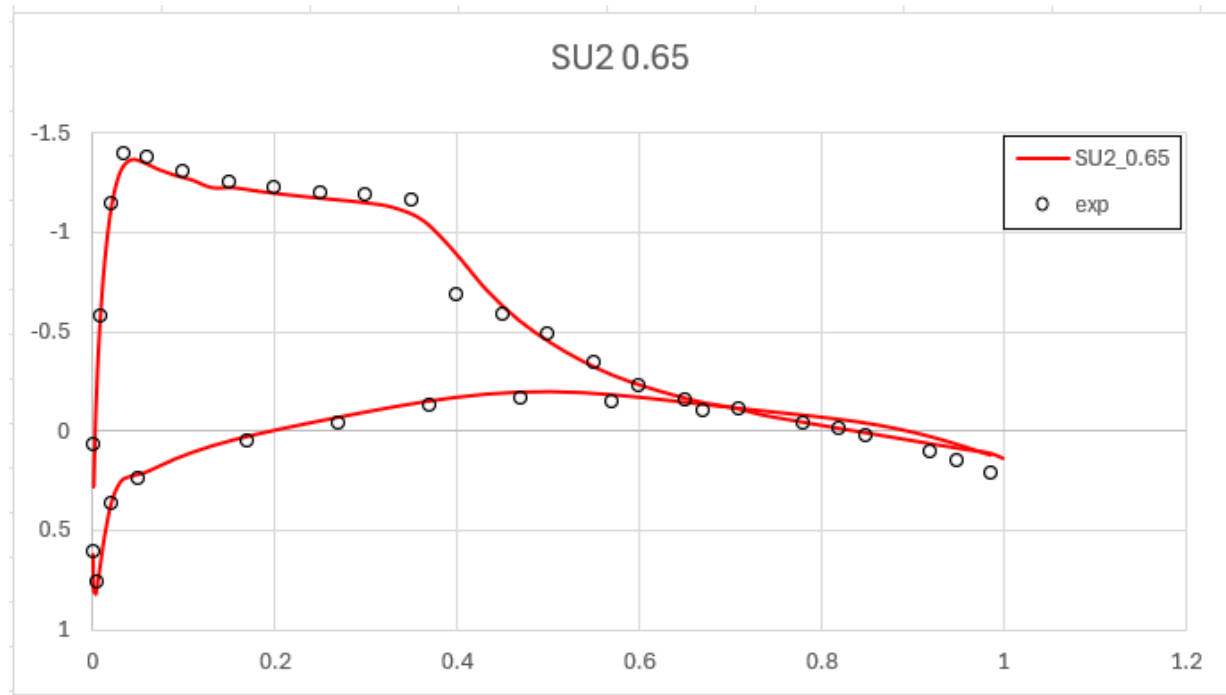
- 우측과 같이 값을 뽑은 뒤 엑셀을 통해 열어줍니다.
- 열린 엑셀에서 Point0과 Pressure_Coefficient만 남긴 채 모두 지워주며 이 두가지로 그래프를 생성해줍니다.
- 그래프 생성을 완성하게 되면 겹치거나 꼬이는 값을 재설정해준 뒤 nasa case_2565의 결과값과 비교하여 비교 그래프를 그려줍니다.



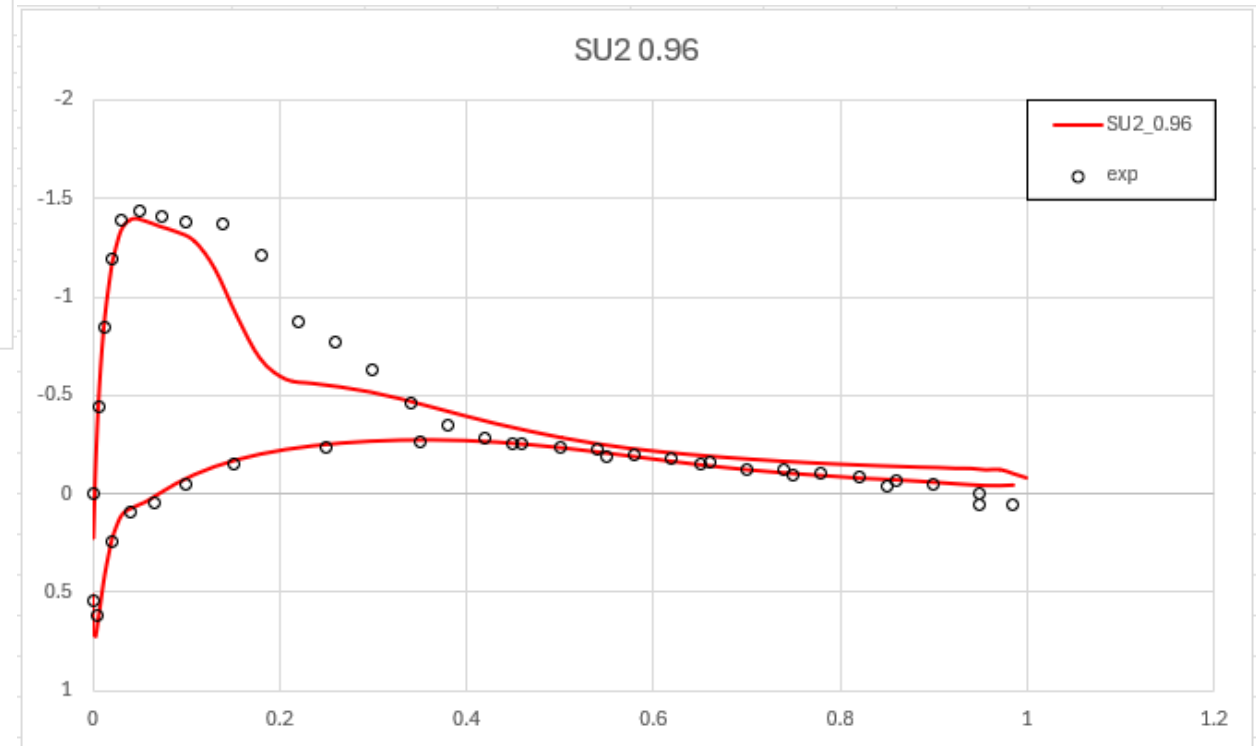
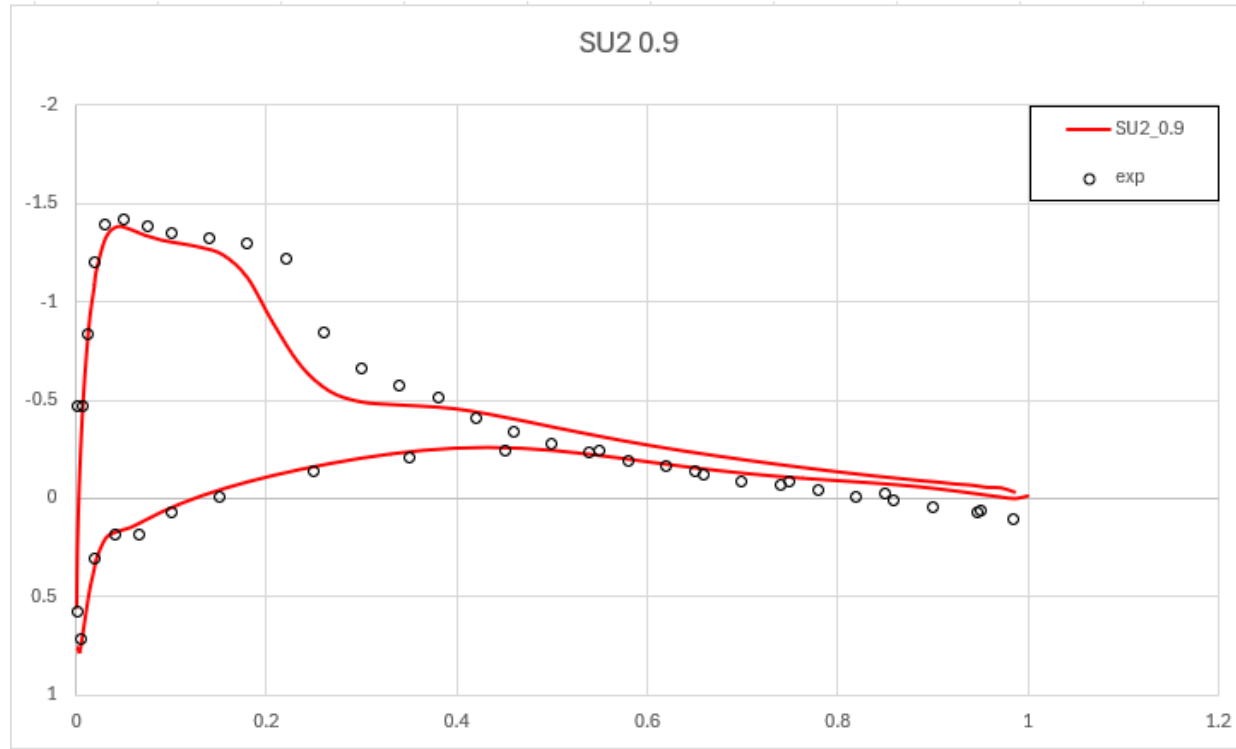
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